

THE WALL STREET CLUB

QUANTITATIVE FINANCE INDUCTIONS '26, ROUND 1

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Dataset A: Alpha Corp

Metric computed: Rolling mean, Standard deviation, Z- score

Metric: mean = \$100.36, standard deviation = 1.99

Extreme Z-scores: day 15 = +2.43, day 17 = -2.64

Match: Mean Reversion

Reason: Day 15 has a Z-score of **+2.43**, which is above +2 and indicates the price is unusually high—potentially panic buying/overbuying. Day 17 has a Z-score of **-2.64**, which is below -2 and indicates the price is unusually low—potentially panic selling/overselling. Since both extreme prices return near the mean afterward, Alpha Corp is suited to a mean-reversion strategy.

Dataset B: Beta Dynamics

Metric computed: 3-day vs 7-day simple moving average.

Metric: final 3-day SMA = \$71.83, final 7-day SMA = \$68.43

SMA spread: \$3.40

Match: Trend Following (Momentum)

Reason: The 3-day SMA is **\$3.40 higher** than the 7-day SMA because the most recent prices are consistently higher than the earlier prices included in the seven-day period. This shows recent upward movement is stronger than the broader recent average, confirming positive momentum and making trend following the suitable strategy.

Dataset C: Gamma Tech

Metric computed: Daily range (High-Low) and look at Volume over time.

Metric: day 15 range = \$0.04, volume = 9,000; day 16 range = \$4.50, volume = 2,500,000

Match: Volatility Breakout

Reason: On day 15, the daily range shrinks to its historic minimum of **\$0.04** and volume falls to **9,000 shares**. This can happen when traders are uncertain and activity dries up: buyers and sellers are closely balanced, so price remains tightly compressed.

On day 16, the range jumps to **\$4.50** and volume rises to **2.5 million shares**. A likely explanation is that the upward move crossed stop-loss levels held by short sellers, forcing them to buy back shares to close their positions. That buying can amplify the breakout and push the price higher.

Dataset D: Delta & Echo

Metric computed: mathematical spread (Delta- Echo)

Metric: normal spread = \$10.00, peak on day 15 = \$24.00, final spread = \$10.00

Match: Pairs Trading (Statistical Arbitrage)

Reason: The normal spread between Delta and Echo is **\$10**, but it widens to **\$24 on day 15**. At this point, traders could short the over-performing asset, Delta, and buy the under-performing asset, Echo, expecting their price relationship to return to normal. As the spread later reverts to **\$10**, this supports the pairs-trading strategy: the trade profits if Delta falls relative to Echo, Echo rises relative to Delta, or both occur.

Dataset E: Omega Resources

Metric computed: standard deviation, moving average trends, and daily range consistency.

Metric: No stable mean, standard deviation, sustained momentum, or volatility-compression pattern.

Past prices carry zero predictive information about future prices.

Match: No edge (random-walk trap)

Reason: The high dispersion and inconsistent SMA relationship do not show a stable mean, persistent trend, or repeatable signal; therefore, it has no reliable exploitable edge.